

AI Job Alert Bot using n8n and Telegram

Domain

Career Automation | Agentic AI | Workflow Automation | Job Search Assistance

Problem Description

Students and freshers spend significant time searching multiple job portals every day. Relevant opportunities are often missed because job postings are scattered across different platforms. Manual job searching is repetitive and inefficient.

Objective

Develop an automated AI-powered job alert system that collects relevant AI/ML and data-related jobs, filters them, and delivers daily notifications directly to Telegram.

Proposed Solution

The system uses n8n workflow automation to fetch jobs from RemoteOK, filter relevant roles such as Python Developer, Machine Learning Engineer, Data Analyst, Data Engineer, SQL Developer, and MLOps Engineer, format the results, and send them automatically through a Telegram Bot.

System Workflow

1. Schedule Trigger starts the workflow daily. 2. HTTP Request fetches job data from RemoteOK API. 3. JavaScript Filter identifies relevant AI/ML and data roles. 4. JavaScript Formatter creates a readable job alert message. 5. Telegram Bot sends job alerts to the user. 6. Workflow is published and scheduled for automated execution.

n8n Workflow Components

• Schedule Trigger • HTTP Request Node • JavaScript Filtering Node • JavaScript Message Formatting Node • Telegram Send Message Node

Expected Outcome

Daily job opportunities delivered automatically to Telegram, reduced manual searching effort, faster access to fresher opportunities, and improved job application efficiency.

Technologies Used

n8n, Docker, Telegram Bot API, RemoteOK API, JavaScript, GitHub

Future Enhancements

LinkedIn integration, Indeed integration, OpenRouter-powered AI summaries, duplicate job detection, resume matching, cloud deployment using GitHub Actions, and personalized recommendations.

Conclusion

The AI Job Alert Bot transforms manual job searching into an automated workflow. By combining APIs, automation, and messaging platforms, the system provides a practical agentic AI solution for students and freshers seeking career opportunities.